DATA SCIENCE INTERNSHIP PROGRAMME

Week 2: Business Understanding Report

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Loan Prediction Dataset

Feature Engineering & Data Preprocessing for Machine Learning

Programme	Data Science Internship Programme
Week	Week 2
Project	Feature Engineering & Data Preprocessing for Machine Learning
Dataset	Loan Prediction Dataset
Target Variable	Loan_Status

1. Business Problem

Loan approval is an important decision for financial institutions because it affects both customer access to credit and the institution's financial risk. Loan applications commonly contain information such as income, credit history, employment status, education, marital status, property area, loan amount, and loan-term information.

When large numbers of applications must be assessed, manual review can be time-consuming and may introduce inconsistency. Financial institutions can therefore use data and predictive analytics to support faster, more consistent, and evidence-based lending decisions.

The Loan Prediction dataset contains historical loan applications and their approval outcomes. The business problem is to prepare these records so that a machine-learning model can learn patterns associated with loan approval and subsequently predict whether a new loan application is likely to be approved.

2. Project Objective

The objective of this project is to transform the raw Loan Prediction dataset into a clean, structured, and machine-learning-ready dataset that can be used to predict loan approval outcomes.

The Week 2 workflow focuses on identifying useful variables, handling missing and duplicate records, encoding categorical variables, engineering meaningful features, detecting and treating outliers, analysing correlations, selecting an appropriate feature set, and scaling continuous numerical variables where appropriate.

A key engineered feature is TotalIncome, which combines ApplicantIncome and CoapplicantIncome to represent the total income associated with an application. Correlation analysis showed a strong relationship between ApplicantIncome and TotalIncome (approximately 0.80). ApplicantIncome was therefore removed to reduce redundancy while retaining TotalIncome.

3. Target Variable

The target variable is Loan_Status. It represents the historical outcome of each loan application and is the variable that a future classification model will attempt to predict.

Loan_Status was converted into a numerical binary representation during preprocessing. The target was retained throughout feature selection and final validation because it is the outcome the predictive model is intended to learn.

4. Importance of Predictive Modelling

Predictive modelling can help financial institutions use historical application data to identify patterns associated with loan approval. A trained classification model can provide a consistent prediction based on characteristics learned from previous applications, supporting rather than automatically replacing responsible human decision-making.

Credit_History had the strongest individual relationship with Loan_Status in the analysis, with a correlation of approximately 0.54. This indicates that credit history contains important information for distinguishing between approval outcomes in this dataset.

Correlation alone does not determine whether a feature is useful. Variables with weak individual linear correlations may still contribute through nonlinear relationships or interactions. Feature selection therefore considered both statistical evidence and business relevance.

5. Expected Business Impact

A successful loan prediction model could reduce the time required to process applications by providing an automated first-level assessment. Faster initial assessment could improve customer experience and allow staff to focus on cases requiring additional review.

Predictive modelling can also improve consistency in the initial evaluation of applications by applying learned criteria across records. This can support operational efficiency when application volumes are high.

Better use of historical data may also support risk management by identifying patterns associated with previous loan outcomes. A model should, however, be treated as a decision-support tool and monitored for accuracy, fairness, and appropriate business or regulatory controls.

Finally, an effective predictive system could improve scalability by helping financial institutions process larger numbers of applications without increasing manual review effort at the same rate.

6. Business Questions Addressed

Which features are most relevant?

Credit_History showed the strongest individual relationship with Loan_Status. Other applicant, financial, credit, and property variables were retained where they provided relevant information.

Which variables require encoding?

Categorical variables such as Gender, Married, Education, Self_Employed, and Property_Area required numerical encoding.

Which variables require scaling?

CoapplicantIncome, LoanAmount, and TotalIncome were standardized because their numerical magnitudes differ and scaling is useful for algorithms sensitive to feature scale.

Are there redundant or highly correlated features?

ApplicantIncome and TotalIncome showed a strong correlation of approximately 0.80. ApplicantIncome was removed while TotalIncome was retained as the combined income measure.

How should missing values and outliers be handled?

Missing values were assessed and handled during preprocessing. Potential outliers in continuous financial variables were detected using the IQR method and treated through IQR capping. Loan_Amount_Term was excluded from continuous outlier capping because it represents discrete loan-term categories.

What preprocessing techniques improve dataset quality?

Missing-value handling, duplicate removal, data-type validation, feature engineering, categorical encoding, outlier treatment, feature selection, and scaling improved the consistency and suitability of the data for modelling.

Is the dataset ready for machine learning?

Yes. Final validation produced 613 rows and 13 columns, with zero missing values, zero duplicate records, no non-numerical columns, and the Loan_Status target variable present.

7. Conclusion

The Week 2 business understanding stage establishes a clear predictive modelling objective: preparing historical loan application data for predicting loan approval outcomes. The dataset contains applicant, financial, credit, and property-related information that can be used to learn patterns associated with Loan_Status.

The preprocessing workflow transformed the raw data into a machine-learning-ready dataset. The final validated dataset contains 613 observations and 13 numerical variables, including the target variable. It contains no missing values, no duplicate records, and no object-type columns.

The analysis also demonstrates the value of feature engineering and feature selection. Credit_History emerged as an important predictive feature, while the strong relationship between ApplicantIncome and TotalIncome supported removing the redundant ApplicantIncome variable. These decisions provide a documented, business-informed foundation for the modelling stage.